



Effects of additive manufacturing on convective heat transfer in 3D-printed micro/mini-channels and fluid passages with micro/mini structures – The review

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ABSTRACT

Additive manufacturing is a novel method of producing three-dimensional products. It allows designers to break boundaries in terms of the complexity of the geometries produced, in many cases reducing the number of parts in the final component and enabling more cost-effective production. Such an advantage of the additive manufacturing process is very attractive, especially for heat transfer designs where cooling efficiency can be boosted by developing complex fluid flow paths, and the design space of possible solutions is extended significantly. As 3D printing advances as a manufacturing method, it is crucial to understand its inherent limitations, especially in developing micro-channel structures. This paper presents a comprehensive review of the literature on heat transfer and flow phenomena in 3D-printed micro/mini-channels and fluid passages with micro/mini structures, with a particular focus on surface roughness. This has an impact on friction factors and, consequently, on pressure drops and, according to the Reynolds analogy, also on heat transfer. It is important to understand how the printing strategy can impact the thermal performance of the components. This review will, therefore, focus on the various heat transfer augmentation features and additive manufacturing factors.

1. Introduction

Additive manufacturing, also known as 3D printing, is an advanced manufacturing technology that creates 3D objects by successive addition of material layers [1]. The advantages and capabilities of additive manufacturing have been utilized in many industries as it allows for the development of new products with significantly improved properties, functionalities, and efficiencies at reduced production costs. Advances in this production technology and the recent explosion of materials and methods for developing additively manufactured devices have opened a new era in thermal management [2]. Novel and miniaturized additively manufactured heat exchangers, as well as cooling and heating systems with 3D-printed parts, are finding applications in many industries, e.g., in electronics, automotive, aerospace, gas turbine, medicine, construction, etc. [2]. The 3D printing technology allows engineers to expand the realm of spatial and surface design significantly. This technology demonstrated great improvement in production process automation, a high level of design freedom, the ability to manufacture complex structures, the great potential for customization, optimization, and material savings, and the ability for fast prototyping. Therefore, in the

novel design of heat exchangers and cooling/heating systems, the optimization of convective heat transfer, and significant miniaturization and compactness of products, can be achieved. In these products, functional fluid channels may exhibit very sophisticated shapes and flow path arrangements, which lead to highly extended heat transfer area and high intensification (turbulisation) of fluid flow [2]. Furthermore, 3D printing may simplify the manufacturing process of products with complex, internal fluid channels, which are frequently produced from many parts, into one component. In consequence, in such products with sophisticated fluid channels, which can operate at high-pressure ratios, leakages at the joints can be eliminated, improving the overall performance of the part while maintaining safety and service life. These complex, compact, and efficient products produced with additive manufacturing technology might not be easily obtained by applying traditional manufacturing techniques.

Additive manufacturing technology has a lot of undoubted advantages. However, there are still many challenges in designing and manufacturing 3D-printed products with functional fluid channels and streamlined bodies, especially at the mini and micro scale [3]. These challenges are mainly related to process repeatability and quality. In 3D

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printing technology, it is not easy to precisely control the quality of the formed surface, which will be in contact with a fluid. This includes surface roughness and contour accuracy, which greatly influence the fluid flow structure and heat transfer characteristics [3], especially for smaller hydraulic diameters. On the one hand, a rough surface can augment heat transfer by a trip of a boundary layer in convective heat transport. But on the other hand, it leads to an increase of the pressure drop (loss) across the dissipation region. Therefore, the 3D-printing may provide both opportunities related to increasing heat transfer efficiency and threads associated with channel cross-section shape accuracy and an increase in the pressure drop along the flow path.

The 3D printing technology shows much promise for use in many industries. However, understanding the fluid flow and heat transfer phenomena in parts made with additive manufacturing is crucial for successfully utilizing this technology for certain applications. To date, the majority of research and development efforts in the field of additively manufactured heat exchangers or components of cooling/heating systems have been oriented toward achieving superior thermal performance over devices produced by means of traditional manufactured technologies [4–6]. These researches mainly focused on developing sophisticated geometries, which lead to improved heat transfer performance. However, there are also several studies in which the effects of internal microstructure and surface morphology of additively manufactured parts are studied regarding their influence on the heat transfer and fluid flow phenomena. Some of these studies showed that selected 3D printing process parameters, e.g., building direction or laser scanning parameters, greatly influence surface roughness and related heat transfer. Moreover, investigations were conducted that quantify and qualify the influence of additive manufacturing process parameters and related internal microstructure and surface morphology on the fluid flow and heat transfer.

The goal of this review is to focus on a basic research summary for understanding the limitations of additive manufacturing technology in aspects of repeatability of the process, an appropriate representation of manufactured geometries, and parameters, which can influence the accuracy of the process in terms of roughness and its impact to convective heat transfer and pressure drop. Such knowledge is crucial for developing 3D-printed components for efficient thermal management in many devices and systems.

2. Surface roughness and shape characterization in additively manufactured elements

Incompressible flows characterization and the impact of surface roughness was the area of research from the beginning of the 20th century. In the literature, many theoretical models exist that allow the prediction of friction factors based on relative surface roughness. The friction factor is important in assessing pressure drop in manufactured channels. For the first time, the term relative surface roughness was used by Kohlrusch [7], who defined the ratio of wall roughness to the hydraulic diameter of the channel. The surface roughness of the wall is known as the average value of roughness (R_a) and can be obtained by means, e.g., profilometry measurement. Extensive experiments were conducted by Nikuradse [8], who tried to characterize flow for a broad range of Reynolds (Re) numbers from 600 to 10^6 . Nikuradse experiments were conducted on pipes with a diameter on a macroscale from 25 to 100 mm. The main conclusion was that roughnesses do not impact the friction factor in the laminar and transition regime of the flow. This experimental work is dated to the 1933 year. However, it should be noted that Nikuradse [8] did not perform any heat transfer measurements in sand roughened tubes. Researchers came back to this topic at the end of the 20th century and started investigating microscale (channels from 100 to 1000 μm range) and see if the theory developed by Nikuradse is still valid. For example, a summary of new information can be found in [9–12]. Firstly, the rise in the friction factor relates to an increase in the surface roughness. Additionally, researchers observed

that the transition zone between laminar and turbulent flow, could appear earlier with increased surface roughness. These findings are very important to 3D-printing technology. All the benefits that additive manufacturing can bring to engineers, especially the desire for an increase in cooling effectiveness of designed parts in many situations by means of using microscale cooling features, make the topic of surface topology characterization and understanding extremely important to understand the limitations of this manufacturing method better.

One of the most popular methods for additive manufacturing is the Direct Metal Laser Sintering (DMLS), which belongs to the family of Powder Bed Fusion (PBF) methods. The production process in PBF methods starts by supplying powder from which the part will be manufactured into the bed, which is moving in the radial direction. Next, the laser provides a high heat flux in a direction based on a digital model of the part, which sinters powder in the intended shape. After solidifying the first layer, the bed drops and refills the amount of powder using a recoater, and the process starts again until the whole part is finished. In this method, obvious parameters influencing the final part shape and, as a consequence, surface roughness can be the size of the bed movement (layer thickness), laser beam spot size and energy, and powder particle size.

It was observed that 3D printing process parameters greatly influence surface morphology and related surface roughness. Two primary modes for roughness generation were identified [13]. They are the staircase effect and the dross formation at downward-facing surfaces due to gravity forces, as is shown in Fig. 1. The staircase effect corresponds to layer-by-layer additive manufacturing practices and is directly related to the layer height and the discrete partitioning of the slicing conducted prior to the manufacturing process [13].

The effect of additive manufacturing parameters on surface morphology was investigated by Fox et al. [14]. They built samples covering a range of overhang angles and process parameters in the Laser Powder Bed Fusion (LPBF) system. Analysis was then performed to determine a relationship between process parameters, i.e., beam power, beam velocity, overhang angle, and surface roughness. In turn, Klingaa et al. [13] carried out a complete roughness characterization of circular 17–4 PH stainless steel channels manufactured by the LPBF. Samples were produced with different global and local building orientation angles. The investigation confirmed that the roughness was highly dependent on both the global and the local orientation.

Studies related to the control of the boundary slope of optimized parts in density-based topology optimization for additive manufacturing were carried out by Wang et al. [15]. The boundary slope of a component affects the amount of support structure required during its fabrication by additive processes and is directly related to the resulting surface roughness, which in turn affects the heat transfer efficiency. By constraining the minimal boundary slope, support structures can be eliminated or reduced, while by constraining the maximal boundary slope, high surface roughness can be attained, and thus, the heat transfer efficiency may be increased. The proposed control strategy was applied to various examples, including linear elasticity, heat transfer, and thermoelastic problems. Experimentally printed test coupons confirmed the ability of the proposed strategy to surface roughness control of the printed parts.

Oter et al. [16] investigated the impact of placing the additively manufactured parts on the build platform. They concluded that location and laser angles connected to this location influence the final surface roughness of the parts. They also observed that the amount of support structures has an impact on the morphology of the surface, which can relate to the intensity of heat transfer processes (i.e., increased heat conductive effect through support structures) during part manufacturing.

Snyder et al. [17] studied build direction on the shape of the micro-channels. During the investigation, three build directions were investigated: vertical, horizontal, and angled (i.e., 0° , 90° , and 45° angles to the build direction). The conclusion was that the unsupported surface of the

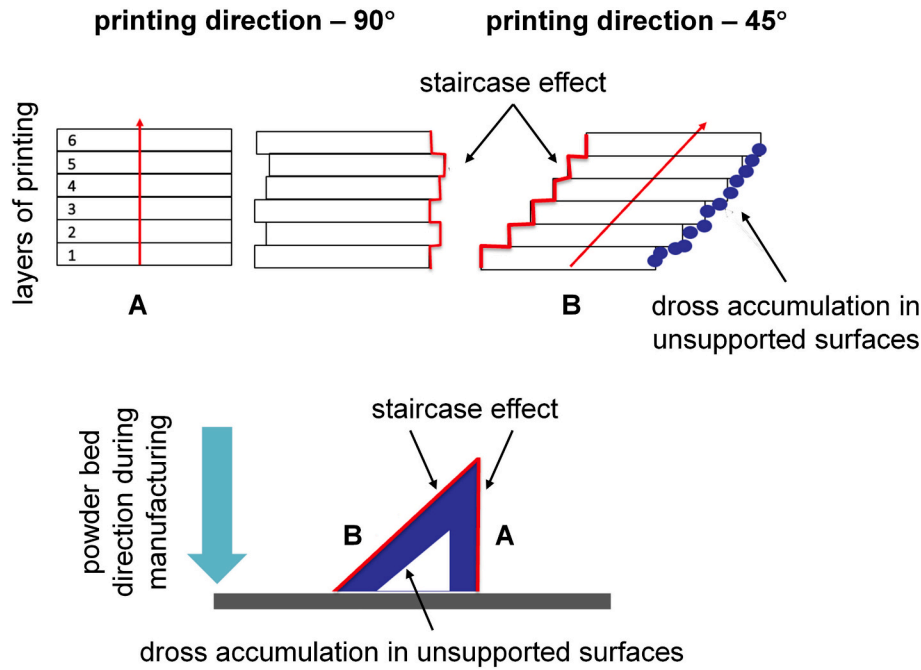


Fig. 1. Example of staircase effect and cross accumulation on surface roughness (where: A – printing direction of 90° , B – printing direction of 45°).

part was always deformed and had the highest friction factor value. Shape modification can be implemented during the design phase to compensate for these deformations. Such a check was done by Snyder et al. [18]. Very similar results were obtained by Wildgoose et al. [19]. During this study, five buildup angles were checked, i.e., 0° , 30° , 45° , 60° , and 90° , and three diameters of cylindrical channels (i.e., 750, 1000, and 1250 μm). For coupons with angles below 60° , the shape of the cylindrical section of the channel was deformed, regardless of the size of the channel. For 60° and 90° , the coupons' surface roughness changed minimally regardless of the diameter of the channel. In terms of surface roughness, it increased with the decrease in buildup angle. Wildgoose et al. [20] studied the impact of build layer thickness and wall thickness of the channel on the surface roughness of the produced part. During the study, a couple of turbine airfoil geometries were printed with varying wall thicknesses and build layer thicknesses (i.e., 40 and 80 μm). The main conclusions were that for thinner walls, especially below 600 μm , the surface roughness started to rise significantly. Additionally, the surface roughness of printed elements was higher for thicker building layers, which corresponds to the staircase effect during the manufacturing process (see Fig. 1).

This quick review shows the importance of knowing factors that can impact additive manufactured design by engineers. When deciding on additive manufacturing, the designer should plan out a strategy for choosing the appropriate angle of the print and incorporate corrections for shape change.

3. Convective heat transfer and friction factors characteristics in additively manufactured micro/mini channels

In the area of convective heat transfer analysis in the 3D-printed micro/mini-channels and around 3D-printed micro/mini streamlined bodies, few studies were carried out in terms of the roughness or different additive manufacturing process effects parameters on pressure losses and heat transfer augmentation. The first paper, which investigated the effect of roughness on convective heat transfer in the 3D-printed flat and finned heat sinks, was published by Ventola et al. [21]. They studied external flow over fins manufactured with the DMLS. They found that the roughness significantly augments the heat transfer on rough flat surfaces, i.e., peaks of 73 % and 63 % on average were

observed for the convective heat transfer enhancement compared to smooth surfaces. These results refer to Reynolds numbers (on heated edge) within the range of 3500–16,500. One of the first papers devoted to the examination of the effect of build direction on the quality of internal micro-channels (i.e., the hydraulic diameter of 508 μm) and resulting friction factor and heat transfer was published by Snyder et al. [18]. In this work, test coupons made of Inconel 718 were produced by applying the DMLS process, more generally referred to as Selective Laser Melting (SLM). Three different build directions were studied (vertical, 45° , and horizontal), with the Reynolds number ranging from 200 up to 12,000. Flow and heat transfer tests revealed a significant influence of build direction on pressure loss, while additive manufacturing process parameters did not significantly affect thermal performance. It was found that vertically built coupons had the lowest friction factor, while the diagonally built coupons had the highest friction factor. Similar investigations but for rectangular micro/mini-channels were performed by Stimpson et al. [22] with the only effect of roughness investigated, while 3D printing process parameters were not studied. The Reynolds number range examined was from approx. 1000 up to 50,000. While high augmentations in flow intensity resulted in increased pressure losses, the Nusselt number augmentations did not linearly scale with the friction factors increase. In addition, both works [18,22] demonstrated that correlations for friction factor, e.g., Nikuradse [8], and heat transfer coefficient, e.g., Dittus-Boelter, for smooth and rough surfaces no longer hold true for additively manufactured surfaces with high roughness values. In the next study, Stimpson et al. [23] focused on characterizing the morphology and quantifying the roughness of micro-channel surfaces made with the DMLS. They developed correlations that relate the roughness measurements to the flow friction and heat transfer for Reynolds numbers up to 50,000, which gives better predictions than correlations for a smooth surface.

Parent-Simard et al. [24] presented experimental and analytical studies of the influence of surface roughness on the performances of a 3D-printed, counterflow, mini-channel heat exchanger. Results showed that the tolerances of the process, which was at $\pm 100 \mu\text{m}$, reduced the mini-channels' hydraulic diameter (due to roughness and shape inaccuracy) and increased the pressure drop. They recommended adjusting additive manufacturing process parameters to tighten the process tolerances for cases where important dimension deviations severely affect

the design's hydraulic and thermal performance. Recently, Snyder and Thole [25] investigated opportunities to tailor additively manufactured surfaces through control of the process parameters to enhance pressure loss and heat transfer characteristics in the internal and external cooling geometries applied in the turbine systems. Test coupons with mini/micro-channels were additively manufactured by applying the LPBF with different laser powers, scan speeds, and scanning strategies and were examined in the Reynolds number range of 1000–12,000. Experimental results showed that the process parameters significantly impact the surface morphology, leading to differences in cooling performance. Moreover, the research showed the great potential of manipulating the parameters within the additive manufacturing process to control the surface morphology, directly influencing the fluid flow and heat transfer characteristics. In a recent paper, Flys et al. [26] evaluated the effect of surface roughness and channel dimensions on the pressure drop and heat transfer in the conventional cooling channels produced by the PBF using the laser beam melting additive manufacturing technique. Compared to the previously mentioned papers, Flys et al. [26] consider conventional channels of a quite high diameter equal to approx. 10 mm instead of mini/micro-channels with diameters smaller than 1000 μm . In the first step of the research, cooling channels with as-build surfaces were compared to post-processed cooling channels such as extrude honing and drilled channels, respectively, considering flows with Reynolds numbers up to 100,000 and water as a working medium. In the second step, the influence of the surface topology on the pressure drops and heat transfer was investigated by applying typical parameters characterizing the surface. They concluded the research that the complexity of surface topology of as-build channels needs to be described by parameters suitable for the detection of fluid interaction. They also suggested further examination of the combination of effects of different manufacturing process parameters on pressure drop and heat transfer.

Simple circular channels printed on a microscale are not the only interest of research groups. As mentioned above, additive manufacturing allows for overcoming the limitations of the shape of the channels, which cannot be achieved with conventional manufacturing techniques. An example of such a study is the work of Wildgoose et al. [27]. Nine channels with different shapes (i.e., circular, hexagonal, pentagonal, elliptical, diamond, square, rectangular, trapezoidal, and triangular cross-sections) were manufactured using the DMLS process from Inco 718 material and the same build direction (90° angle). To compare cooling efficiency and quantify the impact of additive manufacturing, the same cross-sectional perimeter was chosen, and the hydraulic diameter of all channels varied from around 1000 to 1500 μm . Pressure drop and heat transfer tests were performed in a broad range of Reynolds numbers from 500 up to 50,000. The paper showed that all coupons were manufactured with a bigger cross-sectional area regardless of the channel shape. Perimeters were closer to the design intent. Friction factors compared to the smooth channel for the range of Reynolds number varied from 2 up to 4 times, and heat transfer augmentation also compared to smooth channel using Dittus-Boelter equation were higher from 20 to 70 % (see Fig. 2). Rectangle, square, triangle, and trapezoid shapes had the most significant variation of surface roughness which corresponded to friction factors and in consequence Nusselt number increase. From a performance standpoint, the pentagonal shape was the best compared to a circular channel, reaching heat transfer close to the circular channel but with lower pressure drops. Of course, designers can trade pressure drop with the desire to maximize heat transfer if the requirements of the part allow for such a situation.

An additional example of using the benefit of additive manufacturing is the work of Kirsch et al. [28], where micro-channels in sinusoidal shape were studied. Three configurations of wavy channels with wavelength (λ) compared to the length of the coupon (i.e., $\lambda = 0.1 L$, $0.2 L$, and $0.4 L$ – see Fig. 3) and diameter of around 550 μm were manufactured using the DMLS process and INCO718 material. Tested Reynolds regimes were from 1000 to 10,000. Coupons with longer

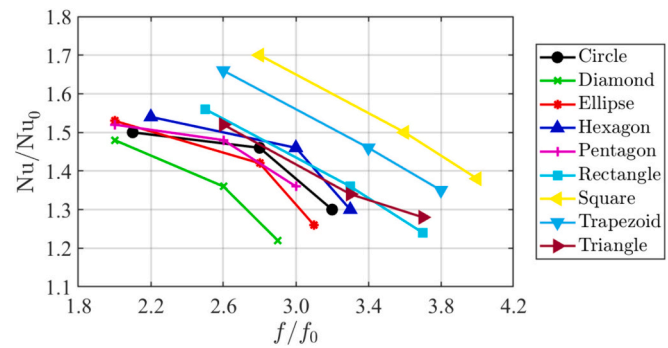


Fig. 2. Nusselt numbers and friction factors compared to the smooth circular channel for different cross-sectional shapes of micro-channels over a range of fully turbulent Reynolds numbers [27] (where Nu_0 and f_0 – Nusslet number and friction factor for the smooth circular channel, and Nu and f – Nusslet number and friction factor for the additively manufactured channel).

wavelengths (i.e., $\lambda = 0.2 L$ and $0.4 L$) performed significantly better compared to shorter wavelength coupons (i.e., $\lambda = 0.1 L$). Friction factors for short wavelength coupons were consistently higher; however, heat transfer augmentation reached the same levels as shown in Fig. 4. Significant heat transfer boost compared to smooth straight channel with the same diameter calculated using the Dittus-Boelter equation was observed on all parts below Reynolds number equal 4000. The authors concluded that more studies should be performed to optimize channels from a cooling efficiency perspective and consider additive process parameters.

Other types of 3D-printed mini/micro-channels are those that act as fluid passages with micro/mini pin fins located within the channels, additionally turbulating the flow and providing extra heat transfer area. Such channels were considered in the next paper [31]. It aimed to provide insights into the combined effects of high surface roughness with the complex flow structures characteristic of micro-pin fin arrays, which is a special consideration of channels created by metallic protrusions manufactured using the LPBF. Investigated samples showed extremely high roughness on some surfaces. Moreover, the surface roughness was found to be a function of the manufacturing process and design, i.e., pin spacing in the array and the shape of the pin itself, with more pins in the array resulting in higher surface roughness and more distorted pin shapes. The observed increase in friction factor for Reynolds number up to 20,000 between different test coupons with the same spanwise spacing, but different streamwise spacing, was attributed to the difference in surface roughness levels. When compared to their smooth pin fin array counterparts, additively manufactured arrays showed an increase in friction factor of 20–40 % (for a tighter spanwise spacing) and 60 % (for a wider spanwise spacing) over smooth arrays with the same spanwise spacing manufactured with traditional methods. In turn, the capabilities of the DMLS process to produce an array of micro-pin fins were assessed in [29] to understand how the 3D printing process affects the friction factor and heat transfer performance. Triangular, star, and spherical shaped pin fins placed in micro-channel test coupons were investigated for pressure loss and heat transfer at a range of Reynolds number of approx. 200–20,000. The influence of spacing, number of pin fins in the array, and pin fin geometry were tested. Unfortunately, the diameters of the pins varied, resulting in a change in the number of pins on the test coupon, which could significantly impact friction factors and final heat transfer augmentation, as reported by the research team. The summary of different pins' performance can be seen in Fig. 5. Ferster et al. [29] concluded that more research needs to be conducted to fully understand additively manufactured pin fins, to optimize the types of finned heat exchangers and to find the effects of additively manufacturing process parameters on surface roughness and related micro-pin fin arrays performance.

The ongoing development of additive methods allows more complex

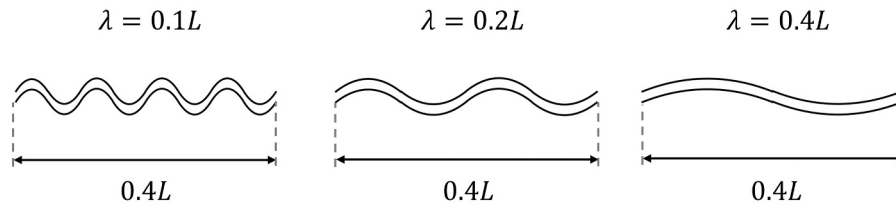


Fig. 3. Visual representation of channel waviness [28] (where: λ – wavelength and L – length of the channel).

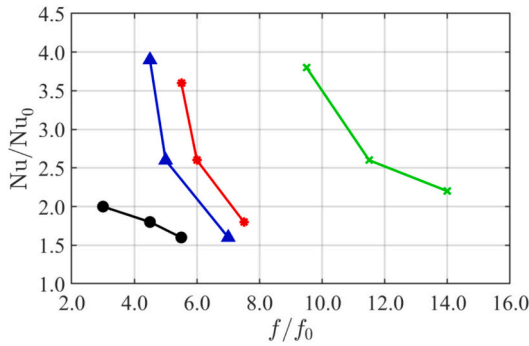


Fig. 4. Friction factor vs. heat transfer augmentation of wavy channels [28].

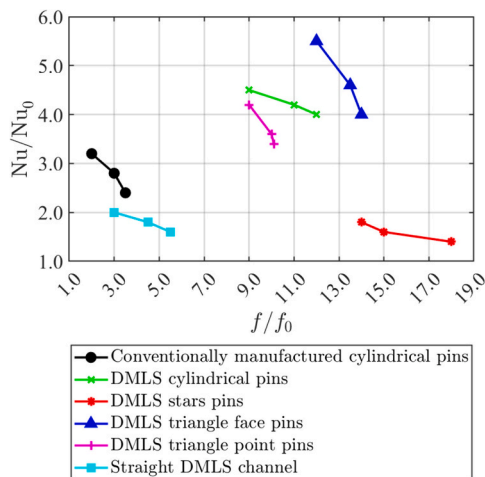


Fig. 5. Friction factor vs. heat transfer augmentation of the DMLS manufactured mini/micro-channels acting as fluid passages with micro/mini pin fins [29]. A conventionally manufactured pin bank was added for reference [30].

structures to be considered that can be used from a thermal performance perspective in novel heat exchangers or cooled turbine components. The research group from the University of Florence presented their results in [32]. The goal of the investigation was to characterize the thermal performance of coupons produced using the Direct Metal Laser Melting (DMLM) process, containing mini-channels acting as fluid passages with complex internal structures. Seven configurations of internal structures were prepared, i.e., pin fin, dimples, simple matrix, double matrix, and three configurations of Kagome lattice structure with different densities (i.e., low, medium, and high). Pin fin configuration is a solid pin staggered configuration. Dimples is a technique of augmentation of heat transfer where, on a targeted surface, valleys with the shape of a hemisphere are placed. Matrix coupon was designed by a combination of 45° turbulators on the bottom surface and on the upper surface negative to 45°, causing turbulators to criss-cross in the middle surface of the coupon, creating a subchannel. Such structures enforce flow going through turbulated channel and suddenly does 90° turn, turbulating

flow structure and increasing heat transfer coefficient. Double matrix configuration incorporates additional rib between upper and lower turbulators, which, in consequence, causes denser subchannels. The last family of configurations is based on the Kagome structure, which is a three-dimensional structure formed by the vertices and edges of different densities. More details of simple and double matrix structures and the Kagome cell can be seen in Fig. 6.

Tests were carried out for turbulent regimes with Reynolds numbers from 15,000 to 35,000. From the measured configurations, the lowest pressure drops were observed for the reference smooth coupon and dimples geometry. It was found that the friction factors were strongly dependent on the porosity of the test specimen and the wetted area. When the area was lower, then the pressure drop was also lower. For the lattice structures, trends showed that the higher the porosity, the higher the friction factors. The highest pressure drop, rising steeply with Reynolds numbers, was reported for double matrix configuration.

According to the Reynolds analogy, heat transfer results are proportional to those representing the pressure drops. Higher average Nusselt number configurations show the dependence on the Reynolds number with a power law with a lower exponent. So here, the lowest heat transfer multiplier compared to the smooth surface was reported for the dimples test coupon and the highest for the double matrix. Fig. 7 shows the friction factor multipliers relative to the smooth mini-channel calculated using the Colebrook equation for appropriate hydraulic diameters vs heat transfer multipliers.

4. Conclusion

Additive manufacturing is a new method that significantly opens design space for engineers, allowing them to create normally non-producible parts using conventional methods. Through the last few years, this method advanced significantly, but still, there are areas where cooperation between the research world and industry needs to be continued. This review summarized parameters of additive manufacturing that can influence cooled part dimensions, surface morphology, surface roughness, and, consequently, their impact on pressure drops and heat transfer augmentation. The main factors are

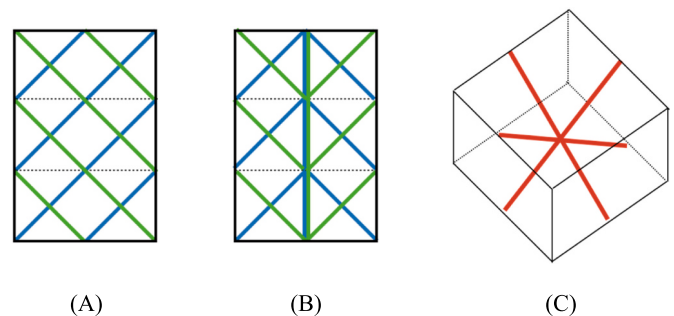


Fig. 6. A) Top view of a simple matrix (blue lines represent the bottom level of turbulators, green top level), B) Top view of the double matrix (blue lines represent the bottom level of turbulators, green top level), and C) Elementary Kagome cell. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

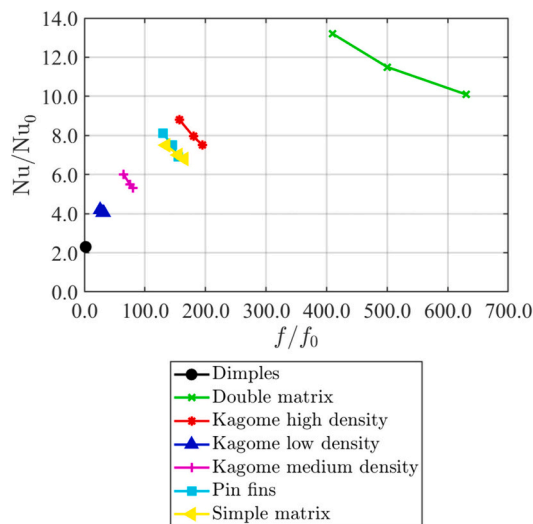


Fig. 7. Friction factor multipliers vs heat transfer enhancement for tested mini-channels acting as fluid passages with complex internal structures [32].

building angle, the thickness of the building layer, and laser parameters, but also some not obvious factors like, e.g., placement of the part on the building plate, amount of support structures, or some specific dimensional relationships (e.g., a wall thickness of the part).

3D printing allows engineers to improve cooling efficiency for components on a microscale, which has been investigated in several works. The conclusion is that it is extremely important during the part's design phase to consider possible shape changes and increased pressure drop due to surface finish to obtain final requirements. This summary discussed mini/micro-channels in different shapes and sizes, providing experts with guidance on what is achievable and what amount of trade needs to be taken.

The current state of the art in terms of materials used to produce additively manufactured parts are alloys based on Co, Cr, and Al. The size of the channels is limited by the capabilities of current machines and is restricted to regions of 500 μm . There is still a niche in information and fluid flow and heat transfer research results for channels smaller than 500 μm – if such channels can even be produced using 3D printing methods, the question arises of what will be relative roughness and heat transfer augmentation? Moreover, can materials based on Cu have the same capability, or due to thermal conductivity production will be penalized with bigger uncertainties? Near years will probably help answer all these questions and allow engineers to produce more thermally efficient products.

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CRediT authorship contribution statement

Tomasz Bulsiewicz: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Piotr Łapka:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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